

Reliability Assessment of JAXA H3 Fairing under Manufacturing Variability

PCE as a Stochastic FEM Surrogate

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Section 1

Motivation

Why Uncertainty Quantification for Rocket Fairings?

JAXA H3 Rocket Fairing

- CFRP skin + Al-Honeycomb core sandwich structure
- Manufacturing scatter is unavoidable:
 - Fiber volume fraction variation
 - Cure cycle fluctuations
 - Adhesive film thickness variation
- How much does this affect structural reliability?

Approach: Non-intrusive PCE as SFEM surrogate
(existing Abaqus model as black-box FEM solver)

Uncertain parameters (5):

Parameter	CoV
E_1 (CFRP modulus)	5%
G_{12} (shear mod.)	10%
K_n (CZM stiffness)	15%
G_{Ic} (fracture energy)	20%
t_n (CZM strength)	15%

Section 2

Methodology

Polynomial Chaos Expansion (PCE)

PCE approximates the model response as:

$$Y \approx \hat{Y} = \sum_{\alpha \in \mathcal{A}} c_{\alpha} \Psi_{\alpha}(\xi)$$

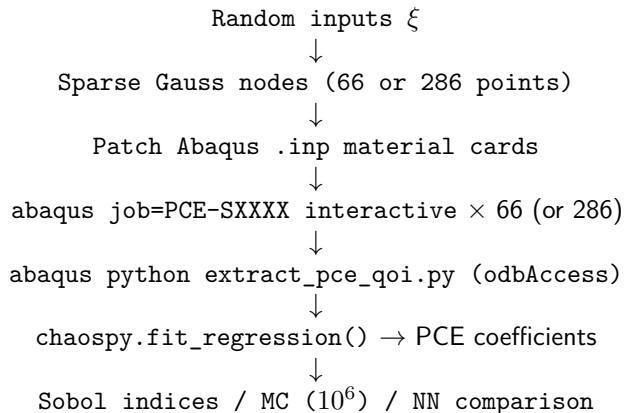
- Ψ_{α} : Hermite polynomial basis (orthogonal w.r.t. Gaussian measure)
- c_{α} : coefficients fitted by least-squares regression
- Non-intrusive: Abaqus treated as a **black box**

Sparse Smolyak quadrature avoids the curse of dimensionality:

Degree p	Full grid	Sparse grid	Reduction
2	$3^5 = 243$	66	$3.7\times$
3	$4^5 = 1024$	286	$3.6\times$

{Negative weights $\Rightarrow$ fallback to `fit_regression` (chaospy)}

Non-Intrusive SFEM Pipeline



Section 3

Results

PCE Statistics: Convergence at Degree 2

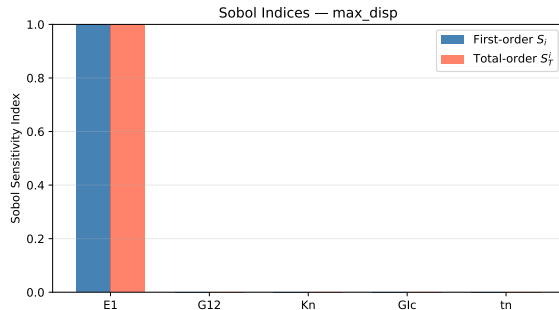
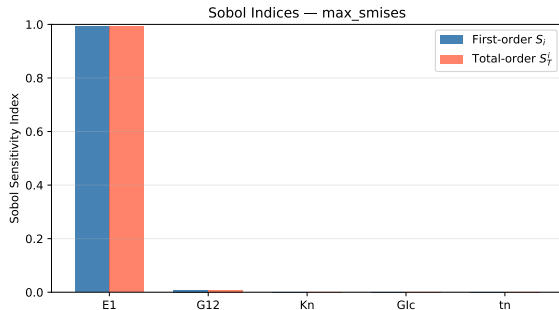
Qol	Degree	Mean	Std	CV	P_f
$\sigma_{vM}^{\max}$	2 (66 pts)	209.7 MPa	2.38 MPa	1.13%	0
$\sigma_{vM}^{\max}$	3 (286 pts)	209.7 MPa	2.38 MPa	1.13%	0
$u^{\max}$	2 (66 pts)	10.07 mm	0.178 mm	1.76%	–
$u^{\max}$	3 (286 pts)	10.07 mm	0.178 mm	1.76%	–

Key result

Statistics identical between degree 2 and 3 $\Rightarrow$ **converged at degree 2**

$P_f = 0$: peak stress 209.7 MPa $\ll$ 1,200 MPa failure threshold

Sobol Sensitivity Indices



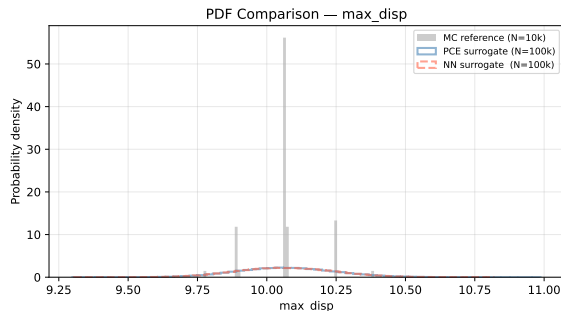
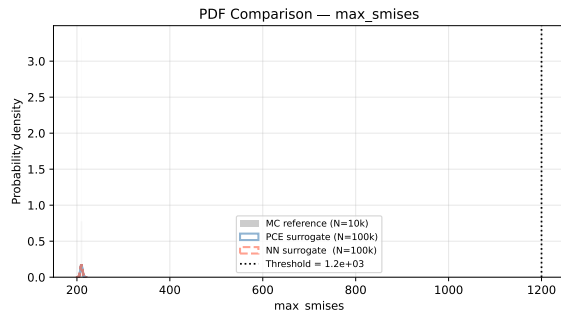
Finding

E_1 dominates **over 99%** of output variance for all Qols.

CZM parameters (K_n , G_{Ic} , t_n): negligible (≈ 0) — adhesive never damages.

No parameter interactions ($S_i \approx S_i^T$).

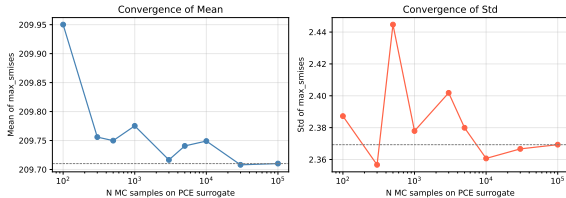
PDF Comparison: PCE vs NN Surrogate



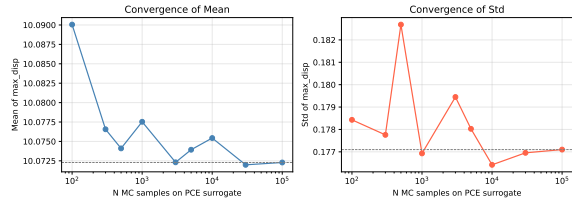
Both PCE and NN surrogates agree with Abaqus samples. \ PCE: analytical PDF via moment propagation. NN: MC sampling required.

PCE Convergence

PCE Surrogate MC Convergence — max_smises



PCE Surrogate MC Convergence — max_disp



Mean and variance converge rapidly. Degree 2 is sufficient.

Degree 2 vs Degree 3: LOO Accuracy

QoI	PCE deg=2	PCE deg=3	NN (66 pts)	NN (286 pts)
σ_{vM}^{max} (MPa)	9.4e-4	1.5e-3	7.2e-4	1.8e-2
u^{max} (mm)	4.2e-5	7.1e-5	2.7e-5	1.6e-3

Counterintuitive NN result

NN accuracy **degrades** with more data (66 $\rightarrow$ 286 points).
Fixed MLP architecture overfits the near-linear response surface.

PCE advantage

Degree 2 PCE is optimal. Degree 3 shows mild over-parameterisation.

Section 4

Discussion

Physical Interpretation

Why does E_1 dominate ($S_{E_1} > 0.99$)?

- Fairing panel: thin-walled under membrane/bending loads
- Response governed by membrane stiffness $E_1 \cdot t$
- E_1 CoV = 5% is relatively small, but it is the *only* stiffness-path parameter
- $\Rightarrow$ Quality control of fiber-direction modulus is the key manufacturing lever

Why is SDEG identically zero?

- Design-point static load — adhesive operates far below cohesive strength t_n
- Even at extreme CoV bounds, interface stress never reaches damage threshold
- CZM sensitivity requires higher load or impact scenario

PCE vs NN: Complementary Tools

Criterion	PCE (this work)	NN surrogate
Data required	66 pts (deg=2)	66–286 pts
LOO accuracy	$\sim 10^{-4}$	10^{-4} to 10^{-2}
Sobol indices	Analytical, exact	Requires separate MC
Interpretability	High	Low (black box)
Uncertainty type	Aleatoric (manufacturing)	Epistemic (model)
Use case	Forward UQ, reliability	SHM, defect detection

Complementarity with existing GNN-SHM work

- GNN-SHM: “Is there a defect?” (epistemic uncertainty via MC Dropout)
- PCE-SFEM: “How does manufacturing scatter affect response?” (aleatoric)
- Independent and complementary — design phase vs. monitoring phase

Section 5

Conclusion

Summary

- ① **PCE degree 2 (66 Abaqus runs) is sufficient** — degree 3 (286 runs) gives identical statistics
- ② **E_1 uncertainty dominates all Qols** ($S_{E_1} > 0.99$)
⇒ CFRP fiber-direction modulus is the key quality control target
- ③ **Structure is reliable** under design load: $P_f = 0$, $\beta = \infty$
(209.7 MPa peak stress vs. 1,200 MPa threshold, CV = 1.13%)
- ④ **PCE outperforms NN** for this smooth, near-linear response
— better accuracy, analytical Sobol indices, no overfitting
- ⑤ **PCE and GNN-SHM are complementary**: aleatoric vs. epistemic uncertainty

Future work

Higher load cases (impact, acoustic fatigue) to activate CZM damage and reveal CZM parameter sensitivity

- ① Ghanem & Spanos (1991). *Stochastic Finite Elements: A Spectral Approach*. Springer.
- ② Sudret (2008). Global sensitivity analysis using PCE. *Rel. Eng. & Sys. Safety*, 93(7).
- ③ Smolyak (1963). Quadrature and interpolation formulas. *Dokl. Akad. Nauk SSSR*.
- ④ Saltelli et al. (2008). *Global Sensitivity Analysis: The Primer*. Wiley.
- ⑤ Dassault Systèmes (2024). *Abaqus Analysis User's Guide*, v2024.

Thank you for your attention.

Questions?